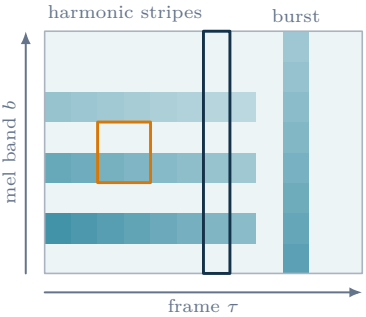
LOG-MEL SPECTROGRAM



orange: $p_\tau \times p_b$ patch → token

navy: column = B-dim frame

NEURAL CODEC + RVQ

encoder: one latent step per ℓ samples

z_1	z_2	z_3	z_4	z_5	z_6

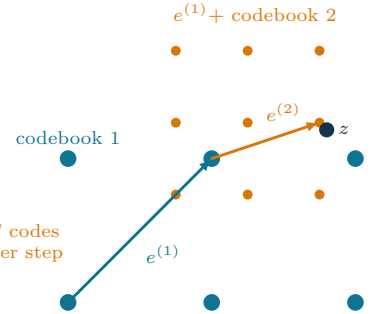
RVQ

s=1	17	903	903	12	640	88
s=2	455	61	302	977	15	230
s=3	90	512	7	349	811	64
s=4	733	28	615	190	402	955

S codes per step

decoder: $\hat{z}_\tau = \sum_s e^{(s)}_{k^{(s)}_\tau}$ → waveform

RVQ IN 2-D



codebook 2 quantizes $r^{(1)} = z - e^{(1)}$

$\hat{z} = e^{(1)} + e^{(2)}$, left over $r^{(2)} = z - \hat{z}$